We are three-dimensional beings, and we live in a curved three-dimensional space. If we were to make measurements of the geometry of our space inside and near a star—the *Schwarzschild geometry*-—we would discover it to be curved in a manner closely analogous to that of the 2D beings' universe.

We can speculate about a higher-dimensional, flat hyperspace in which our curved, three-dimensional space is embedded. It turns out that such a hyperspace must have six dimensions in order to accommodate curved three-dimensional spaces like ours inside itself. (And when we remember that our Universe also has a time dimension, we must think in terms of seven dimensions in all.)

Now, it is even harder for me to visualize our three-dimensional space embedded in a six-dimensional hyperspace than it would be for 2D beings to visualize their two-dimensional space embedded in a three-dimensional hyperspace. However, there is a trick that helps enormously, a trick depicted in Figure 3.3.

Figure 3.3 shows a thought experiment: A thin sheet of material is inserted through a star in its equatorial plane (upper left), so the sheet bisects the star leaving precisely identical halves above and below it. Even though this equatorial sheet looks flat in the picture, it is not really flat. The star's mass warps three-dimensional space inside and around the star in a manner that the upper left picture cannot convey, and that warpage curves the equatorial sheet in a manner the picture does not show. We can discover the sheet's curvature by making geometric measurements on it in our real, physical space, in precisely the same way as the 2D beings make measurements in the two-dimensional space of their universe. Such measurements will reveal that straight lines which are initially parallel cross near the star's center, the circumference of any circle inside or near the star is less than π times its diameter, and the sums of the internal angles of triangles are greater than 180 degrees. The details of these curved-space distortions are predicted by Schwarzschild's solution of Einstein's equation.

To aid in visualizing this Schwarzschild curvature, we, like the 2D beings, can imagine extracting the equatorial sheet from the curved, three-dimensional space of our real Universe, and embedding it in a fictitious, flat, three-dimensional hyperspace (lower right in Figure 3.3). In the